

PH509 Computational Physics
Project Assignment: Numerical Quantum Mechanics
IIT Gandhinagar

This project explores numerical solutions of the Schrödinger equation in one and two spatial dimensions. The goal is to combine physical understanding with careful numerical work. You should write clean, well-documented programs and present clear plots together with a brief discussion of the physics.

Use atomic units unless stated otherwise.

Allowed packages: `numpy`, `scipy`, `matplotlib`. Do not use built-in PDE solvers.

For every problem you must:

- describe the numerical method used,
 - demonstrate convergence with respect to grid spacing, timestep, and domain size,
 - show properly labeled plots,
 - comment on the physics, not only the numerics.
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Problem 1. Time-dependent evolution of a Gaussian wavepacket in 1D

Consider the time-dependent Schrödinger equation

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi.$$

The initial state is a Gaussian wavepacket

$$\psi(x, 0) = \frac{1}{(\sigma\sqrt{\pi})^{1/2}} \exp \left[-\frac{(x - x_0)^2}{2\sigma^2} + ik_0 x \right].$$

Use

$$x \in [-10, 10], \quad x_0 = -3, \quad \sigma = 0.7.$$

First take the harmonic oscillator potential

$$V(x) = \frac{1}{2} x^2.$$

- (a) Discretize space and implement a time-evolution scheme based on the leapfrog method by separating the real and imaginary parts of the wavefunction. Write down the update equations explicitly.

- (b) Verify numerically that the wavefunction remains normalized in time.

Plot

$$N(t) = \int |\psi(x, t)|^2 dx$$

and report the maximum deviation from unity.

- (c) For $k_0 = 0$, evolve the packet long enough to observe at least one full oscillation in the harmonic potential.

Plot

- $|\psi(x, t)|^2$ at several times,
- $\langle x \rangle(t)$,
- the position of the maximum of the probability density.

Compare the motion of $\langle x \rangle$ with the classical trajectory.

- (d) Repeat part (c) for $k_0 = 2$ and discuss the effect of nonzero initial momentum.
- (e) Investigate numerical stability and accuracy by varying the timestep and spatial grid spacing. Comment on the effect on normalization and phase accuracy.
- (f) Replace the potential by a linear potential

$$V(x) = Fx, \quad F = 1.$$

Evolve the packet and compare the motion of $\langle x \rangle(t)$ with the classical uniformly accelerated motion.

Discuss spreading and boundary effects.

Problem 2. Radial Schrödinger equation for the hydrogen atom

In a central potential the Schrödinger equation can be reduced to a one-dimensional radial equation for

$$u(r) = rR(r).$$

In atomic units the radial equation for the Coulomb potential is

$$-\frac{1}{2} \frac{d^2 u}{dr^2} + \left[\frac{l(l+1)}{2r^2} - \frac{1}{r} \right] u = Eu,$$

with boundary conditions

$$u(0) = 0, \quad u(r) \rightarrow 0 \quad (r \rightarrow \infty).$$

Use a finite computational domain $r \in [0, r_{\max}]$.

- (a) Starting from the three-dimensional Schrödinger equation, derive the radial equation and identify the effective potential

$$V_{\text{eff}}(r) = \frac{l(l+1)}{2r^2} - \frac{1}{r}.$$

Plot $V_{\text{eff}}(r)$ for several values of l and comment on the centrifugal barrier and the existence of bound states.

- (b) Using a shooting method or Numerov method, compute the lowest bound-state energies for

- $l = 0$: $1s, 2s, 3s$
- $l = 1$: $2p, 3p$
- $l = 2$: $3d$

Compare your numerical results with the exact hydrogenic formula

$$E_n = -\frac{1}{2n^2}.$$

Using the computed energy levels, estimate a few transition energies between low-lying states and compare with the observed hydrogen spectral lines (see, for example, the NIST Atomic Spectra Database).

- (c) Plot the radial wavefunctions $u_{nl}(r)$ and verify that the number of radial nodes is

$$n - l - 1.$$

- (d) Show numerically that the energy depends only on the principal quantum number n and not on l . In particular verify that

$$E_{2s} \approx E_{2p}, \quad E_{3s} \approx E_{3p} \approx E_{3d}.$$

Explain why this degeneracy is a special property of the Coulomb potential, and comment on how it is reflected in the structure of the hydrogen spectrum.

- (e) Solve the same radial problem using a finite-difference Hamiltonian matrix method. Compare the eigenvalues and eigenfunctions obtained by the two methods and discuss the accuracy and convergence.
- (f) Break the degeneracy by introducing one modification to the Coulomb problem. Choose one of the following:

- impose a hard-wall confinement at $r = r_c$, so that $u(r_c) = 0$,
- replace the Coulomb potential by

$$V(r) = -\frac{1}{r} + \lambda r^2,$$

- replace the Coulomb potential by

$$V(r) = -\frac{e^{-\mu r}}{r}.$$

Compute how the levels with $n = 2$ and $n = 3$ split when the problem is modified. Give a physical explanation of the result and comment on how such a change would modify the corresponding spectral lines.

(g) (Optional: Harder extension)

Compute expectation values such as

$$\langle r \rangle, \quad \langle 1/r \rangle$$

for selected states and compare with known analytic values.

Problem 3. Two-dimensional scattering from a circular potential

Consider the two-dimensional time-dependent Schrödinger equation

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2} \nabla^2 \psi + V(r) \psi, \quad r = \sqrt{x^2 + y^2}.$$

Take a circularly symmetric potential

$$V(r) = \begin{cases} V_0, & r \leq R \\ 0, & r > R. \end{cases}$$

Use a Gaussian wavepacket

$$\psi(x, y, 0) = A \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2} + i(k_x x + k_y y) \right].$$

Suggested parameters

$$(x_0, y_0) = (-8, 0), \quad \sigma = 1, \quad k_x = 2, \quad R = 1.5.$$

As a starting point, you may use

$$x, y \in [-20, 20], \quad \Delta x = \Delta y \approx 0.1.$$

For a split-operator method, $\Delta t \approx 0.003$ is usually adequate, whereas for an explicit finite-difference evolution a smaller timestep, such as $\Delta t \approx 0.001$ – 0.002 , is recommended. You should then refine both the grid and timestep to check convergence.

(a) Describe the numerical method used for time evolution. You may use a split-operator method or a finite-difference scheme.

Explain boundary conditions and normalization.

(b) Set $V_0 = 0$ and show free propagation of the packet.

(c) Use a repulsive barrier ($V_0 > 0$) and evolve long enough for the scattered wave to propagate past the disk.

Plot the probability density at late times and show clearly the interference / diffraction pattern.

Identify

- transmitted wave,
- scattered wave,
- interference fringes,
- shadow region.

(d) Study how the pattern changes when you vary

- the radius R ,
- the momentum k_x ,
- the packet width σ .

(e) Compute an angular distribution

$$P(\theta) = \int |\psi(r, \theta)|^2 r dr$$

far from the scatterer.

Interpret this as a numerical differential cross section.

(f) Repeat for an attractive potential ($V_0 < 0$) and compare the patterns.

(g) Perform convergence tests with respect to grid spacing, timestep, and domain size.

(h) Compare with the classical expectation.

For classical particles incident on a hard disk, trajectories with impact parameter $|b| < R$ are blocked, producing a geometric shadow behind the disk.

Plot your numerical result and identify

- the classical shadow region,
- diffraction into the shadow,
- interference fringes,
- forward scattering peak.

Explain why the quantum result differs from the classical picture.

Submission

Submit

- report (6–10 pages),
- source code,
- short note on which problem was hardest.

Evaluation will be based on correctness, convergence, clarity of plots, and physical interpretation.